

Muhammad Jawad Hassan

Senior AI/ML & Backend Engineer



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Portfolio Website

Profile Summary

Senior AI/ML and Backend Engineer with over 5 years of experience building production-grade Agentic AI systems, RAG pipelines, and GPU-accelerated processing platforms across healthcare, fintech, and insurance domains. I build asynchronous Python and FastAPI backends and AI services, and I have delivered multiple major AI initiatives from architecture through to production deployment on AWS. My strengths are multi-agent orchestration, LLM tool and agent design, knowledge graphs, distributed processing, and observability.

Skills

Languages & Core

Python, SQL, OOP, Ubuntu Linux, Git, Claude Code

Databases

PostgreSQL, MongoDB, Redis, Neo4j, Pinecone, ChromaDB, Weaviate, MemGraph, Mem0, SQLite, DynamoDB, Google BigQuery

Cloud (AWS)

S3, EC2, VPC, Bedrock, SageMaker, IAM, Cognito, Route 53, SQS, DynamoDB, RDS, Lambda, IAM, CloudWatch, Code Pipeline

AI/ML & LLMs

Agentic AI, LangChain, LangGraph, Advanced RAG, Fine-Tuning, HuggingFace, TensorFlow, Scikit-Learn, LangSmith, Weights & Biases, Prompt Engineering, OpenCV, MLFlow, MCP, A2A, , LangFuse, Ragas

Backend & Infra

RestAPIs, GraphQL, FastAPI, Flask, Django, Celery, Temporal, Docker, Kubernetes, Nginx, Terraform, GitHub, GitHub Actions, Sentry, Kafka, n8n, Zapier

Professional Experience

Senior Software Engineer (AI/ML & Backend), *Devsinc*

04/2024 – Present

- Designed and delivered production-grade AI systems across multiple client initiatives spanning healthcare, fintech, and insurance, including multi-agent platforms, a distributed desktop automation agent, RAG pipelines, and GPU audio intelligence.
- Engineered scalable asynchronous FastAPI backends with Temporal, Celery, Redis, PostgreSQL, and AWS services such as Lambda, and Bedrock, enabling distributed orchestration and production AI workloads.
- Built autonomous multi-agent systems with LangGraph, LangChain, Claude, and LangFuse to automate financial research, client reporting, and enterprise workflows.
- Architected a distributed desktop AI agent platform using Claude Code, MCP, and Windows VM workers to autonomously execute Epic EHR workflows without API or DOM access.

Data Scientist, *Nextbridge Ltd.*

09/2022 – 03/2024

- Built end-to-end machine learning pipelines for data preprocessing, feature engineering, & model training.
- Developed and optimized machine learning models for classification, regression, and clustering using Scikit-Learn, XGBoost, and TensorFlow, achieving roughly 88 percent average accuracy across evaluated datasets.
- Collaborated with a cross-functional team to deploy models and automate data workflows, aligning solutions with business KPIs using Python and SQL.

Machine Learning Engineer, *AxcelerateAI*

05/2021 – 08/2022

- Developed and deployed a real-time object detection and tracking system using YOLO and ByteTrack, achieving roughly 92 percent detection accuracy on live video streams.
- Optimized computer vision models for NVIDIA Jetson Nano edge deployment, reducing model size by roughly 45 percent while maintaining production-level inference performance.

Projects

Agentic Multi-Asset Research and Investment Platform, Linvest21

- Architected an autonomous multi-agent financial research system with LangGraph and LangChain in a supervisor and router topology, coordinating specialist agents for macro research, fund selection, and portfolio construction across equities, fixed income, ETFs, and mutual funds.
- Developed an asynchronous FastAPI backend with multi-tenant Keycloak authentication, per-node retry and fallback fault tolerance, and real-time streaming to a chat frontend over Server-Sent Events.
- Wired a multi-provider LLM abstraction across GPT-4o, Claude Sonnet, Gemini, and Groq with LangSmith tracing, and shipped containerized CI/CD through GitHub Actions to AWS ECS across multiple client tenants.
- Built an automated six-stage portfolio construction pipeline that combines a curated fund security master, live market data, sentiment signals, and quantitative metrics into explainable weighted recommendations across ten prebuilt risk models.

Smart Doc, AI Health-Data Aggregation and Assistant Platform

- Built the Python and FastAPI AI services for a health-technology platform that unifies fragmented patient data from EHRs, wearables, labs, and manual logs into a single structured and queryable record.
- Engineered a provider-agnostic conversational AI layer over AWS Bedrock, OpenAI, and Gemini, using a hierarchical routing system and more than twenty-five domain tools that read and write structured data across PostgreSQL, Neo4j, and Memgraph.
- Designed a three-stage asynchronous document pipeline that extracts clinical concepts and resolves them to ICD-10, RxNorm, and LOINC codes in a Neo4j knowledge graph, and added retrieval-augmented generation over patient documents using Weaviate.

Distributed EHR Desktop Automation Agent, Atlas Health

- Architected a distributed AI agent platform using FastAPI, Temporal, and Windows VM workers that autonomously executes Epic EHR workflows through Claude Code without API or DOM access.
- Built a production-grade MCP server exposing desktop automation tools for screenshot capture, OCR, mouse and keyboard control, and PowerShell execution across Citrix-hosted applications.
- Implemented atomic VM allocation with SQL row locking over a per-tenant MariaDB fleet registry, along with checkpoint-based recovery using SQLite snapshots, enabling reliable task resumption and human oversight.

Enterprise Audio Intelligence and Call Analytics Platform, EcoLink

- Engineered a GPU-accelerated audio intelligence microservice in FastAPI that transcribes and analyzes insurance-agency calls using NVIDIA NeMo speech recognition and speaker diarization on CUDA and Torch.
- Built an asynchronous pipeline with FFmpeg normalization and chunking that reliably processes multi-hour and multi-speaker recordings while running transcription and diarization concurrently for lower latency.
- Integrated AWS Bedrock with Llama 3.3 70B to generate summaries, sentiment and agent-performance scores, and prioritized task lists, with strict JSON validation and two-layer GPU memory protection.
- Packaged it for reproducible GPU deployment with Docker, Docker Compose, and NVIDIA Container Toolkit.

Education

Bachelor's in Computer Science (BSCS)

10/2017 – 08/2021 | Lahore, Pakistan

Government College University

CGPA: 3.73 (Gold Medalist)

Awards & Achievements

Graduating Class Gold Medalist, Government College University, Lahore

2x Quarterly Shining Star Award, Developers Incorporate (Devsinc)

99th Percentile in National IT Examination, The Government of Pakistan

Consistently on Dean's Honor List, Government College University, Lahore